

ROV-Based Underwater Litter Collection Gripper

Design & Development of a Cable-Driven Debris Collection
Mechanism for the BlueROV2 Underwater Platform

Author: Ana Alexia CUC · Supervisor: Conf. dr. ing. Anastasios NATSAKIS ·
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8,000,000+ tons of plastic enter the oceans every year

— Jambeck et al., Science 2015

Shallow seabeds trap debris — surface systems can't reach

Diver cleanup: costly, limited bottom time, safety risks

ROVs can operate repeatedly without human risk

No off-the-shelf gripper exists for BlueROV2 at this scale



Project Specs

Designed for the BlueROV2 Configuration

1.3 m - Max. operating depth

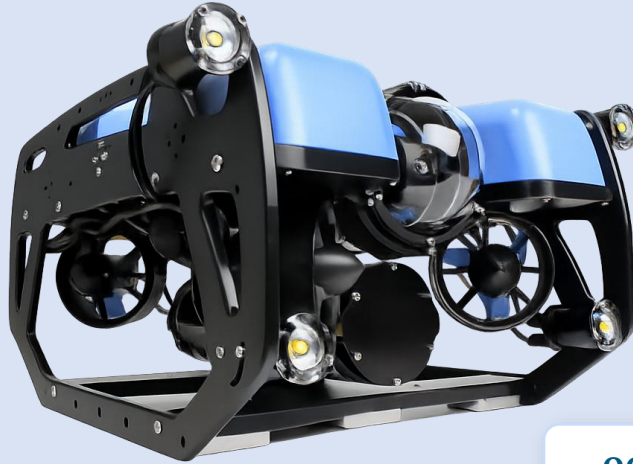
1 kg - Max. weight of the subsystem

10x10x10 cm - Max. litter dimensions

100 g - Max. litter weight (out of water)

Plastic - Target material

Platform – BlueROV2



Raspberry Pi

Onboard computer

Configuration

6 vectored thrusters, ~10 kg

QGroundControl

Topside ground station on operator laptop

Open architecture

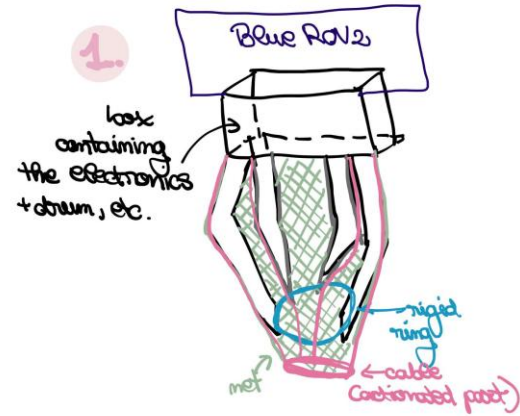
Modular payload support via bottom frame rails

Gripper Mechanism Selection — Decision Matrix

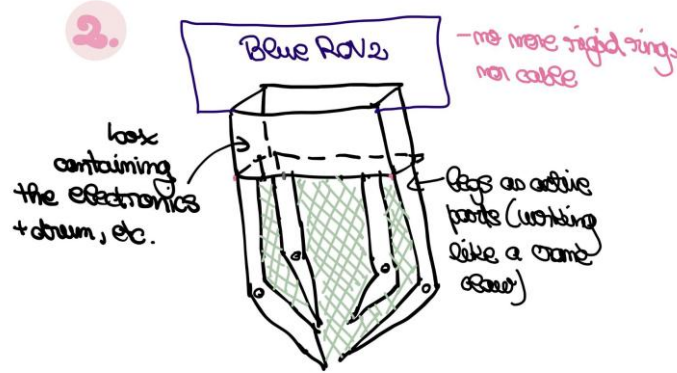
A = Adaptability (0.3) · B = Mechanical Complexity (0.3) · C = Energy Efficiency (0.2) · D = Reliability (0.2)

Mechanism	A	B	C	D	Total
Three-Finger Gripper	2	3	4	3	2.90
Parallel-Jaw Gripper	1	5	4	2	3.00
Suction Scoop / Ramp	2	2	1	2	1.80
Retractable Net	4	4	4	2	3.60
Flexible-Finger Soft	4	2	2	4	3.00
Bellows Soft System	4	2	3	4	3.20
Rigid Basket (petals)	3	3	4	4	3.40

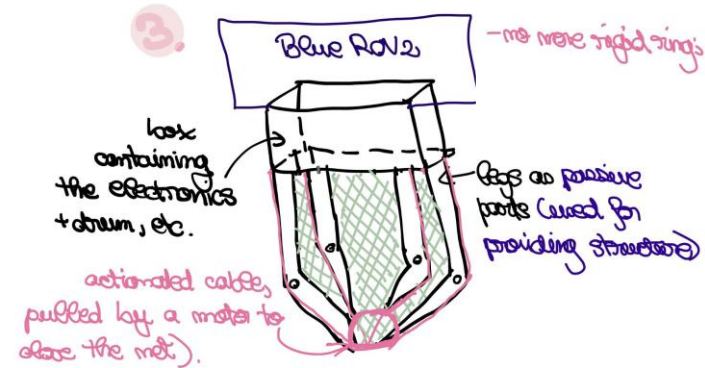
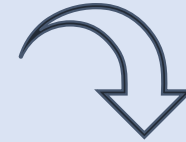
Design Decisions Regarding the Chosen Mechanism



Rigid ring, fixed cable



Active legs (claw-like)



Passive legs, motor-pulled cable – chosen mechanism

Motor, Drum & Coupling

Winding Drum: $\varnothing 70$ mm, 4 cable channels, single 3D-printed part

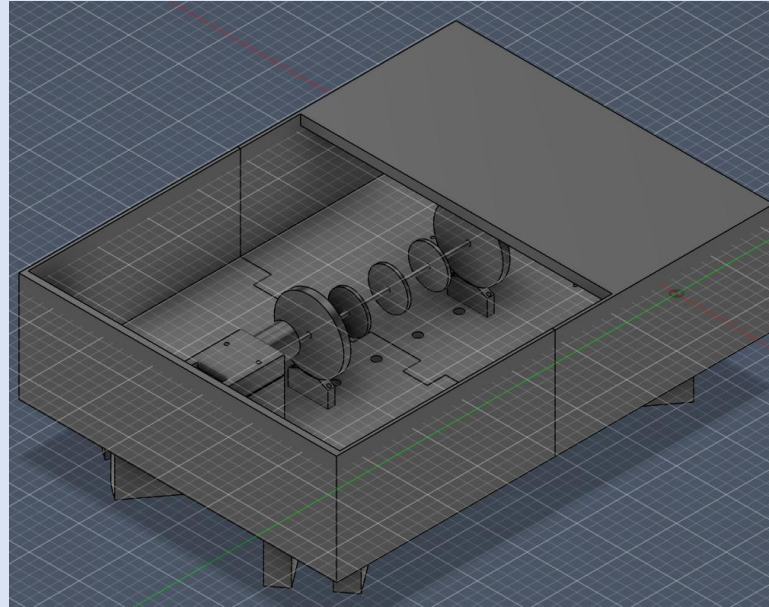
Motor: Pololu 25D, 47:1 @12V, 109 RPM

Encoder: Built-in, 48 CPR feedback

Coupling: $\varnothing 6$ mm each side, matched to motor shaft and drum axle

Cradle: single piece, open-top design

Material: 3D-printed throughout



CAD model of the enclosure

Main Enclosure

Dimensions: 299 × 439 × 180 mm

Material: FDM 3D-printed (PLA)

Split design: 2-part tongue-and-groove joint

Mounting: M3 bolts on ROV transverse bars

Bottom face: 4 cable exits + 8 leg pivot holes

Waterproofing: Partial - future improvement

Gripper Fingers

4

fingers

2

segments
each

M10

pivot bolts

12 N

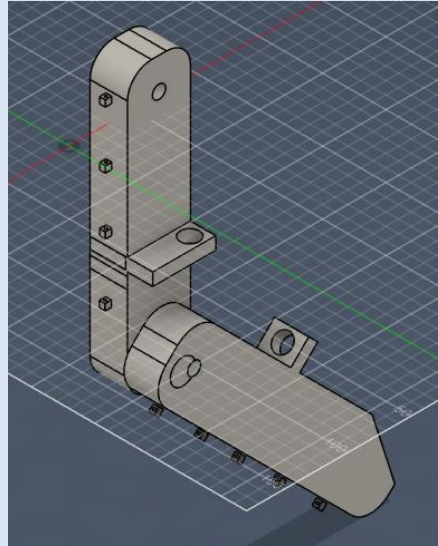
spring force

Configuration: 2 left-hand +
2 right-hand (mirrored pairs)

Actuation: Cable from drum through guide loops

Spring return: $k = 200 \text{ N/m} \cdot dx = 60 \text{ mm} \cdot F = 12 \text{ N}$

Fail-safe: Fingers close passively on power loss



CAD model of the finger



Actual finger

Torque Analysis

1 · Cable Force

$$F_t = \tau_{sp} / l_c$$

35.6 N/leg → 142.4 N total

2 · Ideal Torque

$$\tau_{ideal} = F_t \times r_d$$

142.4 N × 5 mm = 0.712 Nm

3 · Safety Factor

$$\tau_{final} = \tau_{ideal} \times (1 + s_f)$$

× 1.10 design margin

4 · Final Torque

0.783 Nm

within intermittent limit

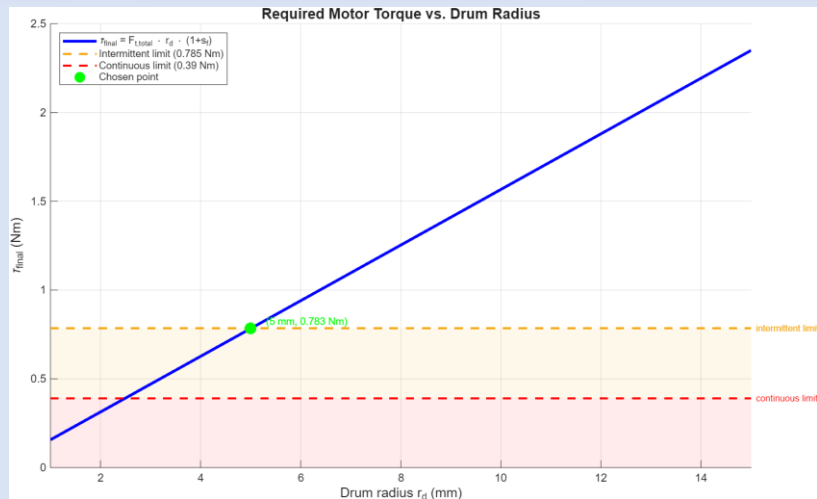
Motor Selected

Pololu 47:1 25D HP 12V

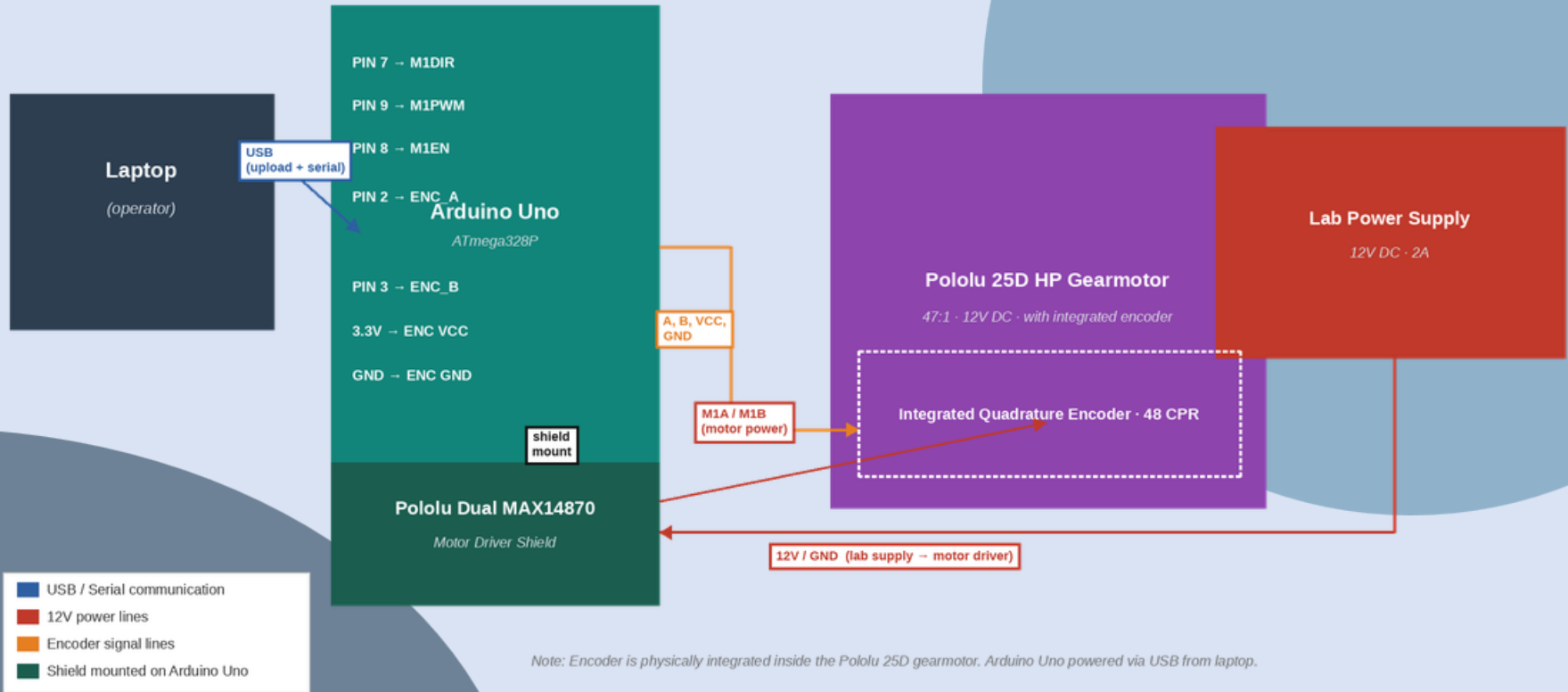
Continuous limit **0.390 Nm**

Intermittent limit **0.785 Nm**

Operating point ✓ 0.783 Nm (≈103 RPM)



Electronic Design



Key Design Decisions



✓ Cable-driven actuation

vs Direct motor per finger

One motor reduces weight, power, and waterproofing complexity

✓ Spring-return (passive close)

vs Active close

Fail-safe: gripper closes on power loss; simpler control logic
Less electric power required

✓ FDM 3D printing

vs CNC machined metal

Rapid iteration, low cost; loads <12 N well within PLA limits

✓ Two-part enclosure split

vs Single-piece box

Printer bed (299×439 mm) limit; tongue-and-groove for self-alignment

Conclusions & Future Work

Contributions

- ✓ Full 3D CAD in Fusion 360
- ✓ Torque analysis
- ✓ Cable-driven gripper with passive spring fail-safe
- ✓ Arduino UNO control architecture with closed-loop position control

Future Work

- Field trials
- Full waterproofing (cable glands, sealed connectors)
- Onboard power (LiPo/Li-ion battery + protection circuit)
- Migration to the ESP32 firmware (Wi-fi/Bluetooth)



Thank you for your attention!

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